Exploring the Effects of Sampling Variability, Scale Variability, and Node Aggregation on the Consistency of Estimated Networks

Arianne Herrera-Bennett[1] and Mijke Rhemtulla[1]

[1]University of California, Davis

Author Note

Arianne Herrera-Bennett. https://orcid.org/0000-0002-3207-4034

Mijke Rhemtulla. https://orcid.org/0000-0003-2572-2424

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Correspondence concerning this article should be addressed to Arianne Herrera-Bennett, Department of Psychology, University of California, Davis, 1 Shields Avenue, Davis, CA, 95616, USA. Email: ariannechr@gmail.com

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Abstract

Work surrounding the replicability and generalizability of network models has increased in recent years, prompting debate on whether network properties can be expected to be consistent across samples. To date, certain methodological practices may have contributed to observed inconsistencies, including use of single-item indicators and non-identical measurement tools. The current study used a resampling approach to disentangle the effects of sampling variability from scale variability when assessing network replicability in empirical data. Additionally, we explored whether consistencies in network characteristics were improved when more items were aggregated to estimate node scores, which we hypothesized should yield more representative measures of latent constructs. Overall, using different scales produced more variability in network properties than using different samples, but these discrepancies were markedly reduced with larger samples and greater node aggregation. Findings underscored the impact of aggregating items when estimating nodes: Multi-item indicators led to denser networks, higher network sensitivity, greater estimates of global strength, and greater levels of consistency in network properties (e.g., edge weights, centrality scores). Taken together, variability in network properties across samples may arise from poor measurement conditions; additionally, variability may reflect properties of the true network model and/or the measurement instrument. All data and syntax are openly available online (https://osf.io/m37q2/).

Keywords: *network models, replicability, generalizability, measurement error, sampling variability*

In recent years, the network analysis approach to multivariate psychology has received increased attention (e.g., see Robinaugh et al., 2020), with growing emphasis on research regarding the replicability and generalizability of network models (e.g., Borsboom et al., 2017; Forbes et al., 2017a, 2017b, 2019; Fried et al., 2018; Funkhouser et al., 2020; Jones et al., 2021; van Borkulo et al., 2017; Williams, 2021, 2022; Williams et al., 2020). Inconsistencies in the literature have prompted debate about the extent to which one can expect network properties to replicate across samples. Although this debate has primarily focused on the psychopathology literature, concerns raised are not exclusive to this domain. Network models have been applied to a range of complex phenomena in order to approximate the structural relations of psychological constructs, including attitudes and beliefs (e.g., Dalege et al., 2017; Rodriguez et al., 2019; Zwicker et al., 2020), cognitive processes (e.g., Golino & Epskamp, 2017; Van Der Maas et al., 2017), classroom dynamics (e.g., Abacioglu et al., 2019; Huitsing & Veenstra, 2012), and clinical disorders (e.g., Borsboom et al., 2017; Fried et al., 2018; Funkhouser et al., 2020). In networks, nodes serve to represent a coherent unit of study (e.g., symptoms, behaviors, facets, or some other variable of interest), whereas edges correspond to the estimated relationships, links, or interactions between these measured units.

Use of different metrics to assess replicability has been discussed as one factor contributing to observed inconsistencies between network models (Forbes et al., 2019). For instance, when assessing the constructs of depression, anxiety, and posttraumatic stress disorder (PTSD), Forbes et al. (2019) found that certain bootstrapping methods (e.g., case-dropping approach; Epskamp et al., 2018) and permutation tests (e.g., invariance tests for comparing global strength and network structure; van Borkulo et al., 2017) each pointed toward the stability and consistency in network properties across samples – in line with conclusions offered by

Borsboom et al. (2017) and Fried et al. (2018). In contrast, when authors tallied the number of edges that were estimated consistently (i.e., present and with the same sign) across pairs of networks, this metric highlighted the differences between networks. In the case of the PTSD data, which consisted of four independent samples each estimating a 16-node network, nearly all edges (i.e., 114 of the 120 possible edges) were estimated as present in at least one of the samples; however, only 39 (or 34.2%) of these were estimated consistently in all four networks. In light of such findings, Forbes et al. (2019) argued that the use of global summary statistics (e.g., total connectivity, global strength scores, or overall correlations between edge weights or node centrality scores) may mask pervasive discrepancies between estimated networks.

Developing and quantifying realistic expectations about consistency in network properties across samples is an important step in evaluating the replicability and generalizability of network models. While some have claimed that inconsistencies in network characteristics between samples undermine the reliability and utility of current network methods and replicability metrics (Forbes et al., 2017a, 2017b, 2019), others have argued that observed differences across empirical datasets “can suggest meaningful differences between samples, random sampling variation, poor reliability in measurement, or a variety of other possibilities” (Jones et al., 2021, p. 5). In direct response to Forbes et al. (2019), Williams (2022) argued that the utility of descriptive methods, such as the tally approach above, will ultimately hinge on the degree to which *observed* replicability rates can be considered or judged in relation to *expected* levels of replicability for any particular metric. With respect to the PTSD example, the number 34.2% should be interpreted relative to the expected percent of consistent edges across four networks randomly sampled from the same population. Without, however, some *expected* baseline against which to compare *observed* rates, making sense of reported replicability indices

becomes much more difficult. To further complicate matters, when it comes to networks specifically, the greater the number of possible edges (e.g., in the case of a larger network), the more unrealistic it would be to expect that all edges replicate. Borrowing from Armour et al.'s (2017) data, Williams (2022) estimated only a 66% probability that more than 34% of the edges in a 20-node PTSD network would replicate.

The network replicability debate underscores the need to understand how different sources of variability contribute to discrepancies in network properties. A core challenge in comparing any set of networks, regardless of domain, is accounting for and distinguishing between factors that introduce noise and variance into the estimation of nodes and edges (e.g., sampling variability, measurement error, contextual variability, population characteristics, etc.). A better understanding of these sources of observed discrepancies should diminish the likelihood that one conflate expected deviations (e.g., due to normal sampling variability) with genuine differences (Jones et al., 2021).

Network Consistency

The current project aims to provide a case study of the differences in network properties that can be expected to arise given the presence of different sources and degrees of real-world variability. While existing work has described the theoretical effects of sampling variability on the expected replicability of partial correlation networks (Williams, 2022), in the present project we used a unique empirical dataset to describe how real differences in items, scales, and samples influence the observed consistency of network results. Our work quantifies the consistency of network models when estimated from empirical personality data and under different measurement conditions. The results can help inform the concepts of network replicability (i.e.,

conditions that explore the effects of sampling variability) and network generalizability (i.e., conditions that explore the effects of scale variability). To date, less work has explicitly focused on assessing network generalizability. One exception is the study by Funkhouser et al. (2020) who explored the generalizability of networks across different populations, and found that network properties were more similar across samples within the same population (i.e., clinical) than across different populations (i.e., clinical and nonclinical).

While the replication literature distinguishes between the concepts of replicability and generalizability, the two are acknowledged as being interlinked. Nosek and Errington (2020) describe replication as a method to confront prior claims with new evidence, whereby consistent evidence increases one's confidence in the original claim, and inconsistent evidence either discredits it or identifies boundary conditions. In light of theoretical immaturity, the generalizability space of a claim is large because there are many conditions (units, settings, etc.) in which it might be supported. In contrast, the more research sees theoretical and methodological progress, leading to theoretical specification and clear predictions, the more the generalizability space shrinks and the replicability space grows.

Our work does not seek to firmly delineate between these concepts; rather, our results represent the extent to which real-world variability in data influences consistency in network properties. For simplicity, we use the term ‘network consistency’ throughout the paper to refer to the similarity in network properties when comparing network models. Because we are specifically interested in the effects of different measurement instruments (i.e. a possible source of systematic variation) on network consistency, our work has implications for the potential boundary conditions arising from choices on how to measure network constructs.

Finally, an added core interest of our study was to explore the degree to which consistencies in network characteristics were improved when a greater number of items were aggregated to measure each node. The networks we examined represent overarching personality dimensions (e.g., Neuroticism), and their nodes represent personality facets (e.g., Anger and Anxiety). Each facet was measured by multiple items on each of the two measurement instruments. While the strong assumptions of classical test theory are unlikely to be met (i.e., the set of items measuring each personality facet are unlikely to be perfectly interchangeable reflections of the latent construct), nonetheless, the items roughly adhere to a factorial structure made up of conceptually distinct facets (e.g., Costa & McCrae, 1992, 1995; for further details, see Methods / Database sections below). Given this measurement structure, we anticipated that aggregating multiple items to create node scores would lead to nodes that are more reflective of the facet-level constructs they represent; the assumption being that if individual items access unique aspects of the facet (e.g., the set of items that measure Anxiety query slightly different contexts), then aggregating them together may produce a more complete and accurate representation of the node, thus increasing its validity. In turn, we expect that networks of more highly-aggregated nodes, versus networks made up of single-item nodes, would be more likely to generalize across samples and instruments. We do note that the extent to which multi-item indicators (vs. single-item indicators) can improve node estimation hinges on theoretical considerations, namely how well one's data meets the assumptions of classical test theory (Lord & Novick, 1968). Aggregating items will work best when the set of items that are aggregated are interchangeable indicators of a unidimensional facet. In that optimal scenario, increasing the number of items that are aggregated to form node scores serves as a proxy for increasing validity and node reliability.

Challenges to Reliable Network Estimation

The literature surrounding network replicability has highlighted a few key challenges. First, given the predominant use of single-item indicators to measure nodes, the issue of poor measurement reliability is particularly relevant in the context of estimating networks, and has been raised as a nontrivial limitation (e.g., Forbes et al., 2017a, 2019; Fried & Cramer, 2017; Funkhouser et al., 2020). In response, some authors have advocated for the use of multiple questions to assess each variable (e.g., the Inventory of Depression and Anxiety Symptoms scale (IDAS); Watson et al., 2007) in order to improve the reliability of node measurements (Fried & Cramer, 2017; Funkhouser et al., 2020).

Second, the use of different instruments or scales to measure the same construct (e.g., Hamilton Rating Scale for Depression (HRSD) vs. Beck Depression Inventory (BDI)) poses conceptual and practical constraints on drawing comparisons across samples: Heterogeneity of scale properties, such as low content overlap (see e.g., Fried, 2017), brings into question the interchangeability of different measures, whereas discrepancies in the number of measured variables necessarily imposes different network structures (Epskamp et al., 2018). These differences in the measurement of network variables limits the extent to which comparisons between networks are interpretable and observed discrepancies between networks informative.

Finally, sampling variability has been raised as an added challenge inherent to – but not limited to – conditional dependence models, such as network models that estimate the pairwise relationships between nodes (i.e., edge weights) via partial correlations (Williams, 2022). Thus, when assessing network replicability, it is important to first assess and quantify the degree to which sampling variability should realistically contribute to discrepancies between network

properties, especially given that effects of sampling variability have been shown to be more pronounced in networks that make use of ordinal data (with fewer categories) as compared to continuous data (Williams, 2022).

Current Study: Overview

To date, the extent to which different sources of variability are expected to individually or jointly contribute to the consistency of network models (or lack thereof) remains unclear. In light of the three limitations identified above, the current study adopted a novel approach to investigating network consistency, specifically by applying a resampling technique to real-world data. In keeping with suggestions from the network literature (see Forbes et al., 2019; Jones et al., 2021), this method allowed us to systematically vary parameters of interest (i.e., sampling variability, scale variability, and node aggregation), whilst maintaining the properties of empirical data. Moreover, what makes our study original is our use of a dataset containing observations from the very same set of participants on multiple instruments (see details below). Previous studies have explored the effects of sampling variability on network replicability, i.e., how networks that are estimated using the same measurement tool compare across different samples or subsamples. The current dataset allowed us to extend this research to include the effects of estimating networks on identical versus independent samples of varying size (i.e., sampling variability), the use of identical versus different scales (henceforth referred to as ‘scale variability’), and the impact of the number of items that were aggregated to measure each node (henceforth referred to as ‘node aggregation’).

With respect to network comparisons, we expected that consistency in network properties would be greater the more similar the samples, and the greater the overlap between scale

characteristics. Moreover, given that smaller samples should produce larger standard errors around network estimates (e.g., node values, edge weights, centrality measures, etc.), we expected that when two networks were estimated on independent samples, discrepancies between observed network properties would decrease under conditions with smaller sampling error (i.e., larger samples) and with more items per node (i.e., greater node aggregation). Similarly, discrepancies between networks, produced as a result of measuring network variables via different measurement tools, are also expected to be attenuated at higher levels of node aggregation. The goal of this study was not to confirm these predictions but to characterize these effects when estimated in real data.

Methods

Database

We made use of the Eugene-Springfield Community Sample (ESCS; Goldberg & Saucier, 2016), an open-source Harvard Dataverse dataset, which consists of data collected across the same sample of individuals, over a period of two decades, on a variety of multi-item personality questionnaires. Included amid these questionnaires are the NEO Personality Inventory-Revised (Costa & McCrae, 1992; from here on referred to as NEO) and the International Personality Item Pool (IPIP; Goldberg, 1999; from here on referred to as IPIP), two widespread instruments used to measure the five overarching personality dimensions (i.e., Neuroticism, Extraversion, Openness, Conscientiousness, and Agreeableness). We selected the NEO and IPIP questionnaires as they seemed particularly well-suited to investigating the effects of sampling variability, scale variability, and node aggregation.

The NEO and IPIP personality questionnaires can be considered two proxies for the same set of constructs: Given the proprietary nature of the NEO instrument, the IPIP questionnaire was specifically developed as an open-source alternative to the NEO questionnaire, and as such was constructed with a similar style of items that shares similar properties (Goldberg, 1999). Specifically, each questionnaire includes five scales to measure the five major personality dimensions; each of these scales is then further subdivided into six subscales to measure the six underlying personality facets of each dimension. The NEO questionnaire has 8 items per facet, and the IPIP 10 items per facet. There is theoretical and empirical evidence that these questionnaires follow a factorial structure (Costa & McCrae, 1992, 1995; Goldberg, 1999). That said, work suggests that this six-factor model does not have simple structure (Johnson, 2014), a conclusion that is backed up by our own analyses (i.e., confirmatory factor analysis demonstrates mediocre fit; see supplementary materials). On the basis of the six-facet structure, we construct six-node personality networks that represent personality dimensions, in which each node represents a single facet of that dimension. Finally, given that both the NEO and IPIP questionnaires are comprised of multi-item subscales to measure each facet, we were able to systematically vary the level of node aggregation by choosing either a single item to represent each facet, or by summing up to eight items.

For simplicity's sake, the current manuscript focuses exclusively on networks modeling the Neuroticism dimension, made up of six facets: Anxiety, Anger, Depression, Self-consciousness, Impulsiveness, and Vulnerability (see Appendix for items and reliability coefficients for each subscale). Reliabilities (omega coefficients) for the Neuroticism facet subscales were comparable across both NEO and IPIP instruments (ranging from 0.74 to 0.90). Mean correlation between facet subscale scores for our sample was 0.77 (ranging from r_p = 0.74

to 0.81). We also ran a replication of our study drawing from the same ESCS database, modeling the Agreeableness dimension, and found the same pattern of results (see supplementary materials).

As personality research goes, network analysis has been adopted to investigate the strength of connections between trait facets (e.g., Conscientiousness facets) and related goals (Costantini et al., 2020), and at different levels of aggregation (i.e., trait-level, facet-level, and item-level; Borsboom et al., 2021). Borsboom and colleagues found that connections between personality variables and personal goals were generally consistent when nodes were treated either as individual items or as broader facets, although certain relations between goals and personality facets were evident at the item level, but not at the facet level. Such findings underscored the idea that items may not all relate in exactly the same way to the same set of variables, suggesting a lack of topological equivalence. These implications are considered in more detail in the Discussion.

Sample Demographics

The ESCS sample consists of participant data collected across a series of questionnaires from 1993 to 2003. In 1993, the sample consisted of 98.4% Caucasian participants (56.9% female and 43.1% male), ranging in age from 18 to 85, and holding varying levels of highest education attained (1.2% not graduated high school, 8.9% high school degree, 54.6% vocational or college degree, 35.2% post-college education; Goldberg, 2018). For the current study, the largest available sample with complete data on both the NEO and IPIP questionnaires were included (N = 424, after excluding cases with missing observations).

Resampling Approach and Experimental Conditions

In order to create a set of resampling conditions for network comparisons, in which different sources of variability were present, we started by sampling pairs of data frames from the full ESCS sample, in three different ways (referred to henceforth as the three "resampling conditions").

I. In the 'sampling variability' resampling condition, we held the measurement tool fixed but varied the sample, that is, we randomly sampled a pair of data frames, where each data frame contained observations from independent subsamples, but were measured using identical scale items. For instance, for single-item indicators, the Anxiety facet could be measured using the same NEO scale item (e.g., "I am not a worrier") in both data frames.

II. In the 'scale variability' resampling condition, we held the sample fixed but varied the measurement instrument, that is, we randomly sampled a pair of data frames, where each data frame contained observations from the same subsample of individuals, but were measured via non-identical items from two different scales. For example, one data frame would measure Anxiety using a NEO scale item (e.g., "I am not a worrier") whereas the other data frame would use an IPIP scale item (e.g., "Am not easily bothered by things").

III. Finally, in the 'sampling and scale variability' resampling condition, we varied both the sample as well as the measurement instrument, that is, we randomly sampled a pair of data frames, where each data frame contained observations from independent subsamples, and were measured using the two different scales.

Additionally, across all three resampling conditions, we introduced five levels of node aggregation by varying the number of items used to estimate each node (i.e., 1, 2, 3, 5, or 8 items). Sets of items (e.g., 3 items per node) were drawn randomly per iteration. While there exist methods that integrate the use of latent variable models and network models (e.g., Epskamp et al., 2017), we simply created sum scores for multi-item indicators prior to estimating the network. Finally, we also manipulated sample size by sampling different proportions of the full sample, namely 20%, 50%, 80%, or 100% of the total number of cases (i.e., n = 84, n = 212, n = 339, or N = 424). In order to avoid overlapping cases between samples, resampling conditions that compared networks across independent samples (i.e., resampling conditions I and III) were necessarily restricted to n's of size 84 or 212.

Taken together – across all three resampling conditions, five node aggregation levels, and two or four sample size levels (depending on the resampling condition)– our design included 40 conditions (i.e., a total of 40 combinations of characteristics for pairs of data frames sampled). For each of the 40 conditions, we generated 50 pairs of networks ("replications"); we limited the design to 50 replications per condition in order to reduce computational burden, which was high due to the network estimation approach.

Network Estimation

For each pair of data frames, the two networks were first estimated separately, then compared on a series of metrics (detailed below). We used the 'ggmModSelect' algorithm (Epskamp et al., 2012) to estimate networks, because it is the default estimation method in the qgraph package that is most commonly used to estimate networks of ordinal data, and it has been shown to have better estimation properties than the EBIC-glasso approach (Williams, Rhemtulla,

Wysocki, & Rast, 2019; Williams & Rast, 2020). The ggmModSelect algorithm begins by estimating a series of 100 networks using the graphical lasso ('glasso'; Friedman et al., 2008) with a range of penalty parameters. These 100 starting networks differ in which edges are estimated to be absent and which are present. Holding these structures (i.e., pattern of edge presence/absence) fixed, each of the 100 networks is re-estimated without the penalty parameter, to obtain non-regularized edge weights. These re-fitted networks are compared using the Bayesian Information Criterion (BIC), and the optimal (lowest BIC) model is selected. Next, starting from this initially optimal model, edges are added or removed, one by one, until the BIC can no longer be improved (Epskamp et al., 2012). The final estimated network (i.e., the weighted adjacency matrix) is a non-regularized network containing a combination of zero and non-zero edges, with non-zero edge weights representing partial correlations. Network edges are thus taken to represent the conditional dependencies between any pair of nodes in the network. All analyses were run in R, using the bootnet package (Epskamp et al., 2018) and the "estimateNetwork" function with default = ggmModSelect and stepwise = TRUE. Across all network estimations, the data were treated as continuous (corMethod = "cor").

Network Comparison Metrics

Consistency between network properties has been assessed within the literature using a variety of metrics. We focused on the set of metrics described below when comparing between pairs of networks across all three resampling conditions. Taken together, these metrics should provide a descriptive overview of the variability in network properties that one might expect given the presence of sampling variability and/or scale variability, and at varying degrees of node aggregation.

While permutation tests, like the Network Comparison Test (NCT, van Borkulo et al., 2017), can be used to statistically test for invariance in network properties, such invariance tests have limited interpretability in our case. Because our resampling approach involved randomly resampling from within the same population sample, this means that across replications, samples drawn will share overlapping observations. As such, results from repeated replications could not be interpreted as a true empirical type-I error rate. Moreover, when varying node aggregation, the larger the number of items being sampled (e.g., multiple-item indicators with 5 or 8 items), the greater the level of overlap in items sampled across replications. For these reasons, when presenting results of our resampling conditions, we focus on reporting descriptive statistics, and interpreting observed trends without appealing to significance tests.

Adjacency Matrices

First, we compared unweighted adjacency matrices, for each pair of networks, by tallying the number of corresponding edges that were estimated as consistent (i.e., absent or present in both networks) or inconsistent (i.e., absent in one network but present in the other). The greater the proportion of consistent edges, the more similar the network skeletons or structures. This metric did not consider the relative size of corresponding edge weights.

Global Strength

Next, we assessed differences in global strength, computed as the sum of all absolute edge weights in the network (van Borkulo et al., 2017). Let V represent the set of nodes in networks (graphs) G_1 and G_2. Let A_1 and A_2 represent the weighted adjacency matrices for G_1 and G_2, respectively. Global strength (for G_1) was defined as

$$S_{G1} = \sum_{i,j \epsilon V} |A_{1ij}|,$$

and difference in global strength (between G_1 and G_2) was defined in terms of distance D

$$D(G_1, G_2) = |S_{G1} - S_{G2}|.$$

Global strength has been used as a measure of overall network connectivity (van Borkulo et al., 2017). The smaller the discrepancy in global strength scores, the greater the similarity in connectivity. Because global strength is computed as an absolute sum score, two networks can yield similar connectivity scores despite differences in network structures and edge directions.

Edge Weight Correlations

Overall similarity between edge weights was assessed with the Pearson correlation (r_p) of the set of corresponding edge weights (i.e., 15 raw partial correlation values) in each pair of networks, averaged across replications.

Centrality Correlations

Overall similarity in node centralities was measured by computing the Spearman-rank correlation (r_s) between the raw Expected Influence scores (Robinaugh et al., 2016) for each pair of networks, and averaged across replications. Expected Influence has been described as a means to identify influential nodes. If one assumes that edges represent equal and bidirectional effects (Borsboom & Cramer, 2013), then Expected Influence accounts for both the activating and deactivating influence a node is expected to have on its immediate neighbors (i.e., the nodes it shares an edge with), and in turn, the rest of the network (Robinaugh et al., 2016). Similar to Strength, a node's Expected Influence is measured as the summed weight of all edges connected to neighboring nodes; unlike Strength, however, Expected Influence distinguishes between positive and negative edges, rather than taking the absolute values of edges. It should be noted that we ran correlation analyses on both Expected Influence and Strength scores; because the

overall trends observed were nearly identical, we only report on the Expected Influence scores within the manuscript (see supplementary materials for Strength scores).

Influence of Instrument Differences on Network Consistency

In order to explore more closely whether we could expect network properties to generalize across different measurement instruments (i.e., NEO vs. IPIP scale), we first estimated and compared whole-sample networks (described below). Next, we drew additional comparisons using a tallied frequency approach, within the 'scale variability' resampling condition, at n = 212 (50% the sample size of the full sample) and at three levels of node aggregation (1-item, 3-item, 8-item).

Whole-sample Networks

Whole-sample networks for the NEO scale and IPIP scale were estimated on the full available sample (N = 424) and the full set of scale items (i.e., 8-item and 10-item indicators, respectively), using the same network estimation approach outlined above. These networks were treated as a best approximation of the population model, and were assumed to yield the most representative reflection of the constituent facets, as operationalized by each of the measurement scales. Comparison between whole-sample networks illustrates the extent to which network properties can be expected to differ when the same construct is estimated using non-identical measurement tools, under the most stable measurement circumstances; that is, when estimated on the same sample of individuals (i.e. no sampling variability), at a sample size twice as large as the observed subsample replications, and using scales with the highest degree of node aggregation.

In addition to descriptive statistics (global strength, number of present and absent edges, density, largest edge weight, most central node) and correlational analyses (edge weights, node centralities), we ran the *NCT* (1,000 iterations) to test the invariance in global strength and edge weights. The *NCT* also includes an invariance test for comparison of network structures. However, given that it operationalizes network structure invariance as the hypothesis that all edge weights are identical across networks, invariance can be violated as a function of one significant edge difference, or alternatively, multiple significant edge differences. The outcome of the network structure invariance test alone, therefore, cannot reveal the full scope of differences that may exist between networks. As such, we only report on specific edge differences when comparing whole-sample networks. Finally, we also computed the *correlation of stability coefficient* (*CS*-coefficient) for each of the networks' Expected Influence estimates, based on 1,000 bootstrapped samples. *CS*-coefficients are meant to serve as an index of centrality stability, and represent the maximum proportion of sampled cases that can be dropped in order to retain a correlation of 0.7 (with 95% probability) between original sample and subsample centrality estimates (Epskamp et al., 2018).

Tallied Frequencies

To illustrate the variability in local network properties, we tallied the frequencies across replications for two edge measures (specifically, the largest edge in each network, and the maximum edge difference observed between pairs of networks) and one node measure (specifically, the most central node in each network). For instance, we assessed the extent to which the same node was ranked as most central by tallying the number of times it was ranked first (across all 50 replications) in the NEO network versus in the IPIP network. Use of these tallied frequencies was meant first and foremost to provide an impression of the expected

variability one can expect when it comes to more local network properties, given the use of different instruments and under varying measurement conditions.

Results

First, we present the results of the three resampling conditions, which explored the individual and joint effects of sampling variability and scale variability, at varying degrees of node aggregation. For simplicity, we only display the effects of sampling variability for networks measured on the NEO scale (all statistics were also computed using the IPIP scale, and trends were comparable). This is followed by a more focused look at the effects of scale variability in order to assess the extent to which network characteristics generalize across instruments.

Effects of Resampling Conditions

Results from our three resampling conditions revealed some clear trends with respect to the effects of larger sample sizes and greater node aggregation on consistency in network properties. Figure 1 demonstrates the results of comparing adjacency matrices across pairs of networks: Plots display the number of edges (max. 15) that were consistently estimated as present or absent (in both networks), or different (in each network). Trends in Figure 1 show that when node aggregation or sample size increased, more edges in the network were estimated as non-zero, indicating potentially greater network sensitivity. While slightly less pronounced, these trends held consistent even when networks were estimated using non-identical items (i.e., in the ‘Scale Variability’ and ‘Sampling & Scale Variability’ resampling conditions). These findings seem fairly intuitive: When nodes were treated at higher levels of aggregation, they may have yielded more representative reflections of the broader facet-level constructs of interest rather than finer item-level behaviors. In turn, resulting networks should be better at capturing and

approximating respective associations between these facets (i.e., edge values) which would otherwise be lost at lower levels of representation (e.g., when appealing to any one item of a subscale). In the same vein, we expected that constructs would be more similarly represented by distinct measurement instruments at greater degrees of node aggregation, and thus would generalize better in multi-item networks, with the caveat, however, that if notable scale-specific differences existed, then these would be expected to persist even when multiple items were used to estimate nodes.

[Insert Figure 1 here]

Figures 2 and 3 display the results of network connectivity as measured by global strength. Figure 2 displays the global strength score, averaged across replications, for NEO networks (left plot) and IPIP networks (right plot), across all levels of sample size and node aggregation. Estimates of global strength increased at higher levels of node aggregation, but were relatively unaffected by sample size (see Figure 2). Increased global strength, under higher degrees of node aggregation, is consistent with the fact that more edges were estimated as present in networks with more items per node (see Figure 1).

[Insert Figure 2 here]

Figure 3 displays differences in estimated network connectivity across pairs of conditions. Overall, differences in global strength were more pronounced at smaller sample sizes and at lower levels of node aggregation; discrepancies were also marginally greater when both sampling and scale variability were present (right plot) as compared to sampling variability (left plot) or scale variability (middle plot) alone. These differences between resampling conditions shrunk with large samples and greater aggregation (e.g., n = 212, 8-item indicators).

[Insert Figure 3 here]

Finally, findings indicated that correspondence between edge weights (Figure 4) and centrality scores (Figure 5) were stronger under more optimal measurement conditions. Figures 4 and 5, respectively, display Pearson and Spearman correlation coefficients, with higher bars representing greater correlation strength. Across both figures, correlation coefficients for 1-item indicator conditions were particularly weak, especially in the 'scale variability' and 'sampling and scale variability' resampling conditions. In general, the overall variability of edge correlations was notably more extreme for conditions with scale variability present, with correlation coefficients spanning a wider range of values (r_p = .09 to .57, and r_p = .17 to .49, at n = 84; see Figure 4, middle and right plots). In contrast, observed edge correlations spanned a relatively smaller range of values in the sampling variability condition (r_p = .30 to .50, at n = 84; see Figure 4, left plot). It is worth reiterating that randomly sampling a single item (or two, or five) from 8 or 10 items could result in the same node being represented by very different items in the scale variability conditions, whereas in the sampling variability only condition, different replications always used the same item(s). These disparities across sampling conditions were attenuated with larger samples (n = 212) and more items per node (e.g., 8-item indicators).

A similar pattern was observed for centrality correlations (see Figure 5, also supplementary Table S5). Additionally, another pattern is evident: Across all three resampling conditions, strength of correlations were comparable when networks were estimated with single-item indicators at n = 212 cases (*r's* = 0.53, 0.47, and 0.31, resp.), *or* when networks were estimated with 2-item indicators at n = 84 cases (*r's* = 0.51, 0.46, and 0.39, resp.). In other words, aggregating across two items (i.e., one additional indicator) yielded the same improvement in network consistency as increasing the sample size by 2.5 times.

These findings highlight the influence of scale variability and node aggregation: If we consider single-indicator networks, greater correspondence in edge weight and node centrality estimates were unsurprisingly observed when networks were modelled using identical items versus non-identical items. Although single items are unlikely to fully represent their intended facets, we would still expect to more systematically observe pairwise associations across two networks modelling the same (vs. different) set of item-specific behaviors or attitudes. This raises an important consideration: The more a construct may rely on a collection of indicators to fully and reliably represent it, the more likely the use of single items will capture only a narrow feature of this broader latent construct. In the case of networks, this may result in item-level networks that have systematically different structures than the facet-level networks that they are meant to represent. This idea resonates with the findings of Borsboom et al. (2021) who observed differences in item-level versus facet-level network structures.

[Insert Figure 4 here]

[Insert Figure 5 here]

Effects of Instrument Differences on Network Consistency

One focal interest of the current study was to afford a characterization of the differences likely to arise when two networks of the same construct are estimated via different measurement scales. First, we estimated whole-sample networks for the NEO and IPIP scales. Whole-sample networks can be assumed to provide a better approximation of the Neuroticism network in the population, but also shed light on scale-specific distinctions in how each instrument captures the facets of Neuroticism. Next, we took a more focused look at the ‘scale variability’ resampling condition and assessed the variability of three local network properties: the largest edge, the

maximum edge difference, and the most central node. Namely, we tallied the frequencies of these outcomes across the subsample networks (i.e., across the 50 replications estimated on *n*'s of 212). This allowed us to evaluate the stability of properties across replications, comparing differences between NEO and IPIP scales, and assess whether this variability was greater under conditions with fewer items per node.

Whole-Sample Networks

Figure 6 displays the whole-sample networks for the NEO and IPIP scales, estimated on $N = 424$ and the full set of items (i.e., 8-item and 10-item indicators, resp.) Overall, assessment of global properties illustrates a strong correspondence between NEO and IPIP whole-sample networks. Differences, however, become more evident when evaluating discrepancies at the level of specific edge and node properties.

Whole-sample Networks: Descriptive Statistics. Table 1 provides a descriptive summary of the network characteristics for each of the whole-sample networks. Differences in adjacency matrices are evident from both Table 1 and Figure 6: Network plots display a greater number of edges estimated as present in the NEO network as compared to the IPIP network (10 edges and 8 edges, resp.). In turn, network density was greater for the NEO scale (0.67) than for the IPIP scale (0.53). Interestingly, despite differences in network densities, measures of global strength were nearly identical across NEO and IPIP networks (2.41 and 2.40, resp.). This is because, on average, the denser NEO network had an overall smaller mean edge strength than the sparser IPIP network (see Table 1). In this way, differences at the level of adjacency matrices did not produce observed discrepancies in network connectivity, that is, when network

connectivity was operationalized as a sum score of all absolute edge weights in the network (van Borkulo et al., 2017).

[Insert Figure 6 here]

[Insert Table 1 here]

[Insert Table 3 here]

[Insert Figure 7 here]

Whole-sample Networks: *NCT* Statistics. To complement descriptive statistics, the *NCT* was run to test for the invariance of global strength, and the invariance of individual edge weights. Unsurprisingly, given the similarity in global strength scores, the *NCT* failed to reject the test for invariance in global strength (difference = 0.01, $p = .731$). With respect to edge differences, edge weights for each of the respective networks are listed in Table 3. The strongest edge was not consistent across both networks. In the NEO network, the edge linking the Depression and Self-consciousness facets was estimated as having the strongest weight; this weight was marginally weaker in the IPIP network (edge weights of 0.35 and 0.31, respectively). In the IPIP network, the strongest edge was between Anxiety and Anger (weight of 0.44). In contrast, this edge weight was markedly smaller in the NEO network (weight of 0.16), and was the second smallest non-zero edge in the entire network. *NCT* was run to test the invariance across all 15 individual edge weights, with Bonferroni adjustment for testing of multiple edges. In the majority of cases, (i.e., 12 out of the 15 network edges), the *NCT* failed to reject the hypothesis of invariance in edge strengths. Three individual edges, however, were significantly different between networks. The greatest difference of edge strengths was between Anxiety and Anger (difference = 0.28, $p < .001$). Additionally, edges linking Depression and Impulsiveness

(difference = 0.24, $p < .001$), and Anger and Depression (difference = 0.23, $p = .03$), were also significantly different. In the case of these latter two edges (i.e., Depression–Impulsivity; Anger–Depression), edges were estimated as present in one network, but absent in the other (see Figure 6 and Table 3). The findings outlined here would imply distinctions in the inferred conditional independence relations among Neuroticism facets, depending on the scale used.

Whole-sample Networks: Correlational Analyses. The correlations between edge weights and centrality scores were strong ($r_p = .74$ and $r_s = .83$, resp); see Figure 7 for visual display of the correspondence between raw centrality scores. Both NEO and IPIP networks yielded *CS*-coefficients of 0.75, demonstrating high levels of stability (see also Figure S2 in supplementary information). Finally, with respect to node properties, rank order of nodes, from most central to least central, was not consistent across the two scales (see Table1): Use of the NEO scale determined Depression as the most central node, whereas use of the IPIP scale ranked Anxiety as most central (see also Figure 7).

Tallied Frequencies

Next, we used a tally approach to better characterize the variability in network properties across replications, when two networks were estimated via different measurement instruments. We tallied the frequencies of three outcomes across all 50 replications in the ‘scale variability’ resampling condition, at $n = 212$, and at three levels of node reliability (1-item, 3-item, and 8-item indicators): i) in each network and for each instrument, which edge was found to be the largest; ii) in each network and for each instrument, which node was found to be most central; and iii) in each pair of networks, which edge was found to be most different. In this way, we explored not only the variability of these specific network characteristics across replications, but

also how discrepancies were potentially diminished when more items were used to measure nodes in the network. Findings support the idea that differences in network characteristics, produced as result of using two distinct scales, are likely to be reduced given greater node aggregation. Moreover, tallied frequencies of edge and node properties underscored how differences across replications may reflect the population network and/or scale-specific differences.

Table 2 summarizes the output from the 'scale variability' condition, displaying average estimates for networks measured via the NEO scale versus the IPIP scale. Visual inspection of Table 2 reaffirms some of the trends already discussed above: At each level of node aggregation, average global strength was fairly comparable between NEO and IPIP networks. As node aggregation increased, average global strength (i.e., sum of absolute edge weights) increased, whereas average edge strength (of non-zero edges) remained fairly stable, reflecting simultaneous increases in the strength of existing edges and the addition of new, weaker edges. Increasing the number of items per node also led to greater overall correspondence in edge weights and centrality scores. Table 2 helps illustrate the merit of using multi-item indicators over single-item indicators. Even a small increase in the number of items used to estimate each node (i.e., from one to three items), led to substantial differences in correlations observed: correlation between edge weights improved from r_p = .25 to .47, and correlation between centrality scores improved from r_s = .47 to .70.

Assessment of local network characteristics (i.e., specific edge and node comparisons) helped underscore the existence of scale-specific differences (Table 2). For instance, for the NEO scale, 29 out of the 50 1-item networks ranked Depression as the most central node, whereas there was almost perfect agreement across 8-item networks (49 out of 50 replications).

In contrast, centrality scores for IPIP networks remained fairly divided even in 8-item networks: 31 out of 50 replications determined Depression as the most central node, whereas 17 out of 50 replications determined it to be Anxiety. These differences between scales are (in retrospect) not surprising given the pattern in centrality scores for each of the whole-sample networks: As visible in Figure 7, the NEO whole-sample network displays a clear rank order when it comes to the Expected Influence of nodes, whereas for the IPIP scale two nodes (Anxiety and Depression) are essentially tied for first. This outcome highlights the role that features of the population network play in replicability outcomes: Variability in network properties, across different samples, should be considered in light of the underlying true network model and scale-specific properties.

A similar pattern was observed for the frequencies of the largest edge measure (Table 2). At the highest level of node aggregation, there was fairly high agreement across IPIP networks that Anxiety-Anger was largest edge (35 out of 50 replications), and a much smaller proportion of replications ranking Anxiety-Depression or Anxiety-Vulnerability as the largest edge (6 out of 50 each). This again reflects the edge weights of the IPIP whole-sample network (see Table 3): The largest edge in the network (Anxiety-Anger = 0.44) was noticeably larger than the second (Anxiety-Vulnerability = 0.36) and third (Anxiety-Depression = 0.35) largest edges. In comparison, NEO network frequencies were more evenly split: Links between Depression and Vulnerability, and between Depression and Self-consciousness, were largest in the majority of replications (26 and 22 out of 50 replications, resp.). This aligns with the NEO whole-sample network in which the two largest edges (Depression-Vulnerability = 0.34 and Depression-Self-consciousness = 0.35) were nearly identical in size.

Taken together, findings imply that when centrality scores or edge weights are similar in size or strength within a network, such as two “equally” influential nodes or two “equally” strong associations between network variables, it should be reasonable to expect greater variability in such metrics across replications. Moreover, whether node or edge properties are observed to be similar in size/strength may importantly be a function of the measurement instrument itself, and how units within the network are conceptualized and operationalized. In other words, variability in network properties across replications may be less reflective of unreliable networks, but rather consistent with properties of the underlying true network model or scale-specific properties.

[Insert Table 2 here]

Discussion

In this paper, we aimed to estimate the degree to which real differences in participants, measurement instruments, and items influence the consistency of network properties across replications. We used a unique empirical personality dataset to investigate these influences under less versus more stable measurement conditions (i.e., larger samples and higher degrees of node aggregation). Our investigation revealed some clear patterns that may shed light on empirical network replication successes and failures.

Node Aggregation

At greater levels of node aggregation, we observed denser networks (i.e., more non-zero edges), a result that may indicate higher network sensitivity. Accordingly, multi-item indicators led to higher estimates of global strength across individual networks, and smaller discrepancies between pairs of networks. In multi-item networks, we also observed greater correlations between edge weights and between centrality scores, which was even noticeable when moving

from single-item indicators to 2- or 3-item indicators. Specifically, going from a single indicator to two improved consistency in centrality scores as much as increasing the sample size by 2.5 times. These findings underscored the influence of node aggregation on the consistency of network properties, and reinforced the idea that aggregating items can be an effective way to gain a more valid representation of higher-level nodes. This finding may help explain the observed low replicability rates for single-indicator networks in the literature, especially if we consider that two scales for the same construct (e.g., depression) can share little content overlap (Fried, 2017).

The effects of item aggregation raise the more general issue of topological structures and topological equivalence between networks. At the outset of the manuscript, we hypothesized that use of multi-item indicators may serve as proxies for increased validity and/or reliability of node estimates (see also Fried & Cramer, 2017; Funkhouser et al., 2020). Under this interpretation, changes in network properties (e.g., increased network density) and observed improvements in replicability (e.g., greater correlations between edge weights and centrality measures) may be attributed to improvements in the precision and generalizability of node estimation. If we assume that the set of items that measure each node are interchangeable measures of a single underlying construct (i.e., that the assumptions of a factor model apply), then such an interpretation is noncontroversial: Reducing the degree of measurement error involved in estimating the variables of a network will lead to more accurate estimates of the connections between these nodes, and in turn a more accurate reflection of the true network model. In the context of network analysis, one can conceptualize this assumption of interchangeable items as topological equivalence across single-item networks. Put differently, if items are exchangeable within a facet, then any network made up of a random set of items (namely, one indicator per node) should, at the population

level, have the same topological structure. If this assumption of interchangeable items or topological equivalence does not hold, then we need to consider the effects of node aggregation in light of alternative explanations. For example, it could be that aggregating items with different topological structures creates blended, heterogeneous nodes that all correlate with each other to some extent; that is, we would expect denser networks with more uniform edge weights when each node consists of a multidimensional variable made up of a mix of items. In most empirical situations, we would expect that a combination of these two phenomena is at play: Items on a subscale do, to some extent, measure the same underlying construct, and as such, aggregating them produces a more reliable indicator of that construct. Items on a subscale may also have unique variance that is uniquely related to other items in the measure, such that aggregating items will lead to murkier, less unidimensional nodes.

Sampling and Scale Variability

Another key take-away of our study is the relative impact of sampling variability versus scale variability when comparing effects across resampling conditions. Network properties were observed to be less consistent or more variable in conditions where scale variability was present. For instance, even at the highest level of node aggregation (8-item indicators, n = 212), edge correlations were still weaker in the ‘scale variability’ condition (r_p = .66) than the ‘sampling variability’ condition (r_p = .71), and weakest when both sampling variability and scale variability were present (r_p = .61). Introducing the use of two different scales to measure the same construct led to added discrepancies in network properties beyond those produced from sampling variability alone. Findings illustrate that how we measure a construct can importantly impact the likelihood that it will generalize in future samples. The impact of distinct measurement tools, however, is expected to be attenuated at higher levels of node aggregation, given that two

different multi-item indicators are expected to tend toward a more similar representation of the same construct than two single items.

Different Measurement Instruments

To assess the effect of different measurement instruments more closely, we evaluated local edge and node properties within the 'scale variability' resampling condition. Overall, observed variability in outcomes across replications (represented as tallied frequencies in Table 2) appeared to be consistent with patterns observed in the whole-sample networks: While, in general, consistency across replications increased with greater levels of node aggregation, not all properties reached consensus across replications. This was especially visible in cases where the two largest edges in a network were roughly equal (i.e., links between Depression and Vulnerability, and Depression and Self-consciousness, in the NEO whole-sample network), or when the two most central nodes in a network were equally influential (i.e., Anxiety and Depression nodes in the IPIP whole-sample network). In these cases, even when networks were based on 8-item indicators (n = 212), the two largest edges or most central nodes were about equally likely to turn up as largest/most central in any given replication. It follows that variability across replications may not necessarily indicate unstable properties, but may stem from properties of the population network that the replications are drawn from. To take an extreme example, if the population network model were characterized by all edges being present and of equal weight, any differences across replications would necessarily reflect only sampling variability. In a similar vein, when measurement instruments are notably different, we would expect observed differences between networks to persist even in larger samples and at higher levels of node aggregation.

To elucidate this point, we can consider more closely elements of the NEO and IPIP Neuroticism scales. While we assumed that the NEO and IPIP scales were viable proxies of the same construct (mean correlation between subscale scores r_p = 0.77), with good levels of reliability for their respective subscales (omega coefficients ranging from .74 to .90), it is possible that these two instruments tapped into different yet real features of the Neuroticism construct and constituent facets. Take, for instance, the Depression subscales which correlated highest amongst the facets (r_p = 0.81), and had the highest reliability coefficients (0.87 and 0.90 for NEO and IPIP, resp.): In the NEO scale, some items may have tapped more into the element of shame or guilt (e.g., "I have sometimes a deep sense of guilt or sinfulness", "I tend to blame myself when anything goes wrong"), whereas the IPIP items may have attempted to capture the idea of self-worth (e.g., "Have a low opinion of myself", "Dislike myself"). Given the complex nature of a facet such as Depression, it is not surprising that content varies between scales or even between items on the same scale (Fried, 2017). Thus, when comparing networks across two different scales, increasing node aggregation may not produce more convergent results in cases where two reliable scales are capturing different aspects of the same construct. Additional research might benefit from comparing network properties of the same construct across different tools, under good measurement conditions (i.e., large samples and multi-item indicators), in order to get a better sense of which network properties are stable regardless of instrument used, versus those which may be more scale-dependent.

Different Metrics

A final point worth addressing is how different replicability metrics allow for different information to be inferred, and may in turn appear to lead to inconsistent results. One example is the observed differences in adjacency matrices for comparisons between whole-sample NEO and

whole-sample IPIP networks (i.e., 10 and 8 non-zero edges, resp.). Despite differences in network densities, measures of global strength were nearly identical across networks (2.41 and 2.40, resp.) due to the fact that the denser network had an overall smaller mean edge strength than the sparser network. In this way, differences regarding which edges were estimated as present or absent did not produce observed discrepancies in network connectivity, that is, when network connectivity was operationalized as a sum score of all absolute edge weights in the network (van Borkulo et al., 2017). Another example is the strong CS-coefficient for both NEO and IPIP whole-sample networks, and the strong correlation between network centrality scores (r_s = .83). Taken together, these metrics would suggest strong stability and correspondence between centrality ranks. What is worth noting, however, is that the high centrality correlation between NEO and IPIP instruments is largely produced by the correspondence in rank order for the three *least* central nodes in the network (i.e., Anger, Self-consciousness, and Impulsivity), rather than the three *most* central nodes (i.e., Depression, Vulnerability, and Anxiety) which are not matched for rank order. If these least central nodes were not matched for rank order, then the correlation between centrality scores would drop from 0.83 to 0.66. Given that there is an interest in the literature in identifying the nodes with the highest, rather than lowest, centrality (e.g., Borsboom & Cramer, 2013; Elliott et al., 2020), and less emphasis on the overall correspondence across all node centralities, then global metrics (such as rank correlations or stability coefficients) may not be as informative as comparisons between specific nodes. These findings resonate in part with concerns raised by Forbes et al. (2019) that use of global summary statistics may mask differences in local characteristics of networks. Ultimately, the more cohesion across conceptualizations and operationalizations of metrics (e.g., how one measures 'network

connectivity'), the clearer the meaning we can draw from observed variability of network properties across samples.

Conclusion

This paper describes the extent to which Neuroticism networks exhibit similar and discrepant properties across empirical samples, as a function of sample size, node aggregation, and scale differences. We view this investigation as descriptive research that aims to characterize the extent to which real variability in participants, scales, and items influences the behavior of a range of local and global network consistency metrics. The empirical basis of this paper raises an important question of generalizability; namely, to what extent can we expect the trends reported here to generalize to other datasets, and other domains of network research? In particular, as we do not have access to the true population properties of the dataset, we are unable to precisely characterize how changes in the population affect replication outcomes. We argue that this weakness of our approach is more than offset by the advantages that empirical data confer. In particular, our findings afforded several conclusions that would not have been possible under a more controlled simulation approach. First, the results highlighted the degree to which differences between two similar but non-identical measurement instruments can produce differences in estimated network properties, independent of sampling variability. In fact, networks estimated on the very same sample of participants but different scales tended to differ more than a pair of networks estimated on two different samples of participants on the same scale. Second, we found large effects of node aggregation, such that networks based on single-item nodes were substantially less stable than those in which nodes were composites of two or more items. This effect appeared to be a function of both increasing reliability (i.e., composites are more reliable than individual items) as well as item-specific properties (i.e., individual

personality items within a facet are not strictly interchangeable). Third, examining the properties of each whole-sample network revealed that the variability in network outcomes across samples can be strongly influenced by the properties of the larger dataset or population from which these samples are drawn. In particular, when a property of the larger network lacks variance (e.g., homogeneous edges or node centrality), network metrics that rely on ordering that variance (e.g., edge correlations, strongest edge, most central nodes) will pick up mostly sampling variance, and lack consistency as a result. We expect that all of these trends will be relevant for understanding the performance of network replicability and consistency estimates across subfields in which variable networks are estimated, even while the precise size of these effects is expected to vary in other domains. In addition, our findings may reveal important lessons for other subfields that differ in systematic ways from the present personality example. For example, psychopathology symptom networks are more likely to have nodes measured by individual items; our findings indicate that these may be the most prone to inconsistent results. While the magnitude of the item-aggregation effects that we find here may not generalize to other datasets (where individual item reliabilities are likely to differ), our finding that these effects are very large (i.e., that aggregating just two items resulted in similar gains as doubling the sample size) suggests that this effect is worth investigating in other kinds of data.

In light of our findings, one practical recommendation that we offer to improve network replicability and generalizability rates is to use sum scores made up of multiple items as nodes. This recommendation is obviously appropriate when the set of items that are aggregated are interchangeable indicators of a unidimensional facet. However, it may also lead to greater consistency when the network nodes are meant to capture more complex or multidimensional constructs, such that individual items carry unique information. Here, we see a potential tradeoff

between replicability and interpretability: More aggregation will result in more stable network analytic results, but may also make it harder to interpret the network as a system of interacting low-level components. Researchers will have to think hard about their goals when choosing an appropriate level of network aggregation.

Disclosure statement

No potential competing interests are reported by the authors.

Data availability statement

The data (i.e., dataset, syntax, supplemental online materials) that support the findings of this study are openly available on the OSF platform (https://osf.io/m37q2/).

References

Abacioglu, C. S., Isvoranu, A.-M., Verkuyten, M., Thijs, J., & Epskamp, S. (2019). Exploring multicultural classroom dynamics: A network analysis. *Journal of School Psychology*, *74*, 90–105. https://doi.org/10.1016/j.jsp.2019.02.003

Armour, C., Fried, E. I., Deserno, M. K., Tsai, J., & Pietrzak, R. H. (2017). A network analysis of DSM-5 posttraumatic stress disorder symptoms and correlates in U.S. military veterans. *Journal of Anxiety Disorders*, *45*, 49–59. https://doi.org/10.1016/j.janxdis.2016.11.008

Borsboom, D., & Cramer, A. O. J. (2013). Network Analysis: An Integrative Approach to the Structure of Psychopathology. *Annual Review of Clinical Psychology*, *9*(1), 91–121. https://doi.org/10.1146/annurev-clinpsy-050212-185608

Borsboom, D., Fried, E. I., Epskamp, S., Waldorp, L. J., van Borkulo, C. D., van der Maas, H. L. J., & Cramer, A. O. J. (2017). False alarm? A comprehensive reanalysis of "Evidence that psychopathology symptom networks have limited replicability" by Forbes, Wright, Markon, and Krueger (2017). *Journal of Abnormal Psychology*, *126*(7), 989–999. https://doi.org/10.1037/abn0000306

Costa, P. T., & McCrae, R. R. (1992). Normal personality assessment in clinical practice: The NEO Personality Inventory. *Psychological Assessment*, *4*(1), 5–13.

Dalege, J., Borsboom, D., van Harreveld, F., Waldorp, L. J., & van der Maas, H. L. J. (2017). Network Structure Explains the Impact of Attitudes on Voting Decisions. *Scientific Reports*, *7*(1), 4909. https://doi.org/10.1038/s41598-017-05048-y

Elliott, H., Jones, P. J., & Schmidt, U. (2020). Central Symptoms Predict Posttreatment Outcomes and Clinical Impairment in Anorexia Nervosa: A Network Analysis. *Clinical Psychological Science*, *8*(1), 139–154. https://doi.org/10.1177/2167702619865958

Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, *50*(1), 195–212. https://doi.org/10.3758/s13428-017-0862-1

Epskamp, S., Cramer, A. O. J., Waldorp, L. J., Schmittmann, V. D., & Borsboom, D. (2012). **qgraph**: Network Visualizations of Relationships in Psychometric Data. *Journal of Statistical Software*, *48*(4). https://doi.org/10.18637/jss.v048.i04

Forbes, M. K., Wright, A. G. C., Markon, K. E., & Krueger, R. F. (2017a). Evidence that psychopathology symptom networks have limited replicability. *Journal of Abnormal Psychology*, *126*(7), 969–988. https://doi.org/10.1037/abn0000276

Forbes, M. K., Wright, A. G. C., Markon, K. E., & Krueger, R. F. (2017b). Further evidence that psychopathology networks have limited replicability and utility: Response to Borsboom et al. And Steinley et al. *Journal of Abnormal Psychology*, *126*(7), 1011–1016. https://doi.org/10.1037/abn0000313

Forbes, M. K., Wright, A. G. C., Markon, K. E., & Krueger, R. F. (2019). Quantifying the Reliability and Replicability of Psychopathology Network Characteristics. *Multivariate Behavioral Research*, 1–19. https://doi.org/10.1080/00273171.2019.1616526

Fried, E. I. (2017). The 52 symptoms of major depression: Lack of content overlap among seven common depression scales. *Journal of Affective Disorders*, *208*, 191–197. https://doi.org/10.1016/j.jad.2016.10.019

Fried, E. I., & Cramer, A. O. J. (2017). Moving Forward: Challenges and Directions for Psychopathological Network Theory and Methodology. *Perspectives on Psychological Science*, *12*(6), 999–1020. https://doi.org/10.1177/1745691617705892

Fried, E. I., Eidhof, M. B., Palic, S., Costantini, G., Huisman-van Dijk, H. M., Bockting, C. L. H., Engelhard, I., Armour, C., Nielsen, A. B. S., & Karstoft, K.-I. (2018). Replicability and Generalizability of Posttraumatic Stress Disorder (PTSD) Networks: A Cross-Cultural Multisite Study of PTSD Symptoms in Four Trauma Patient Samples. *Clinical Psychological Science*, *6*(3), 335–351. https://doi.org/10.1177/2167702617745092

Friedman, J., Hastie, T., & Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, *9*(3), 432–441. https://doi.org/10.1093/biostatistics/kxm045

Funkhouser, C. J., Correa, K. A., Gorka, S. M., Nelson, B. D., Phan, K. L., & Shankman, S. A. (2020). The replicability and generalizability of internalizing symptom networks across five samples. *Journal of Abnormal Psychology*, *129*(2), 191–203. https://doi.org/10.1037/abn0000496

Goldberg, L. R. (1999). A broad-bandwidth, public-domain, personality inventory measuring the lower-level facets of several Five-Factor models. *Personality Psychology in Europe*, *7*(1), 7–28.

Goldberg, L. R. (2018). *(0) Documentation and Sample Demographics*. Harvard Dataverse; V1. https://doi.org/10.7910/DVN/VGMXJX

Goldberg, L. R., & Saucier, G. (2016). The Eugene-Springfield community sample: Information available from the research participants (Tech. Rep. No. 56-1). *Eugene, Oregon: Oregon Research Institute*.

Golino, H. F., & Epskamp, S. (2017). Exploratory graph analysis: A new approach for estimating the number of dimensions in psychological research. *PLOS ONE*, *12*(6), e0174035. https://doi.org/10.1371/journal.pone.0174035

Huitsing, G., & Veenstra, R. (2012). Bullying in Classrooms: Participant Roles From a Social Network Perspective: Social Networks and Participant Roles in Bullying. *Aggressive Behavior*, *38*(6), 494–509. https://doi.org/10.1002/ab.21438

Jones, P. J., Williams, D. R., & McNally, R. J. (2021). Sampling Variability Is Not Nonreplication: A Bayesian Reanalysis of Forbes, Wright, Markon, and Krueger. *Multivariate Behavioral Research*, *56*(2), 249–255. https://doi.org/10.1080/00273171.2020.1797460

Lord, F. M., & Novick, M. R. (1968). *Statistical theories of mental test scores*. Addison-Wesley.

Nosek, B., & Errington, T. (2020). What is Replication. *PLoS Biology*, *18*(3). https://doi.org/e3000691

Robinaugh, D. J., Hoekstra, R. H. A., Toner, E. R., & Borsboom, D. (2020). The network approach to psychopathology: A review of the literature 2008–2018 and an agenda for future research. *Psychological Medicine*, *50*(3), 353–366. https://doi.org/10.1017/S0033291719003404

Robinaugh, D. J., Millner, A. J., & McNally, R. J. (2016). Identifying highly influential nodes in the complicated grief network. *Journal of Abnormal Psychology*, *125*(6), 747–757. https://doi.org/10.1037/abn0000181

Rodriguez, A. L., Stephens, D. P., Brewe, E., Ramarao, I., & Madhivanan, P. (2019). A Network Analysis of Domestic Violence Beliefs Among Young Adults in India. *Journal of Interpersonal Violence*. https://doi.org/10.1177/0886260519889923

van Borkulo, C. D., Bork, R. V., Boschloo, L., Kossakowski, J., Tio, P., Schoevers, R., Borsboom, D., & Waldorp, L. (2017). *Comparing network structures on three aspects: A permutation test*. https://doi.org/10.13140/RG.2.2.29455.38569

van der Maas, H. L. J., Kan, K.-J., Marsman, M., & Stevenson, C. E. (2017). Network Models for Cognitive Development and Intelligence. *Journal of Intelligence*, *5*(2), 16. https://doi.org/10.3390/jintelligence5020016

Watson, D., O'Hara, M. W., Simms, L. J., Kotov, R., Chmielewski, M., McDade-Montez, E. A., Gamez, W., & Stuart, S. (2007). Development and Validation of the Inventory of Depression and Anxiety Symptoms (IDAS). *Psychological Assessment*, *19*(3), 253–268. https://doi.org/10.1037/1040-3590.19.3.253

Williams, D. R. (2021). Bayesian estimation for Gaussian Graphical Models: Structure learning, predictability, and network comparisons. *Multivariate Behavioral Research*, *56*(2), 336-352, https://doi.org/10.1080/00273171.2021.1894412

Williams, D. R. (2022). Learning to live with sampling variability: Expected replicability in partial correlation networks. *Psychological Methods, 27(4)*, 606–621. https://doi.org/10.1037/met0000417

Williams, D. R., & Rast, P. (2020). Back to the basics: Rethinking partial correlation network methodology. British Journal of Mathematical and Statistical Psychology, 73(2), 187-212. https://doi.org/10.1111/bmsp.12173

Williams, D. R., Rast, P., Pericchi, L. R., & Mulder, J. (2020). Comparing Gaussian graphical models with the posterior predictive distribution and Bayesian model selection. *Psychological Methods*, *25*(5), 653–672. https://doi.org/10.1037/met0000254

Williams, D. R., Rhemtulla, M., Wysocki, A. C., & Rast, P. (2019). On nonregularized estimation of psychological networks. Multivariate behavioral research, 54(5), 719-750. https://doi.org/10.1080/00273171.2019.1575716

Zwicker, M. V., Nohlen, H. U., Dalege, J., Gruter, G.-J. M., & van Harreveld, F. (2020). Applying an attitude network approach to consumer behaviour towards plastic. *Journal of Environmental Psychology*, *69*, 101433. https://doi.org/10.1016/j.jenvp.2020.101433

Appendix

NEO instrument (8 items per facet)	IPIP instrument (10 items per facet)
Anxiety (N1) (alpha = 0.84, omega = 0.84)	**Anxiety (N1) (alpha = 0.85, omega = 0.85)**
I am not a worrier. [R]	Worry about things.
I often feel tense and jittery.	Am relaxed most of the time. [R]
I have fewer fears than most people. [R]	Am afraid of many things.
I am easily frightened.	Fear for the worst.
I rarely feel fearful or anxious. [R]	Don't worry about things that have already happened. [R]
Frightening thoughts sometimes come into my head.	Am not easily disturbed by events. [R]
I'm seldom apprehensive about the future. [R]	Adapt easily to new situations. [R]
I often worry about things that might go wrong.	Get stressed out easily.
	Get caught up in my problems.
	Am not easily bothered by things. [R]
Angry Hostility (N2) (alpha = 0.81, omega = 0.81)	**Anger (N2) (alpha = 0.89, omega = 0.90)**
I'm an even-tempered person. [R]	Lose my temper.
Even minor annoyances can be frustrating to me.	Get irritated easily.
It takes a lot to get me mad. [R]	Seldom get mad. [R]
I often get angry at the way people treat me.	Rarely complain. [R]
I often get disgusted with people I have to deal with.	Keep my cool. [R]
At times I have felt bitter and resentful.	Get upset easily.
I am known as hot blooded and quick tempered.	Am often in a bad mood.
I am not considered a touchy or temperamental person. [R]	Am not easily annoyed. [R]
	Rarely get irritated. [R]
	Get angry easily.

Depression (N3) **(alpha = 0.86, omega = 0.87)**	**Depression (N3)** **(alpha = 0.90, omega = 0.90)**
I am seldom sad or depressed. [R]	Seldom feel blue. [R]
I have a low opinion of myself.	Have a low opinion of myself.
Sometimes things look pretty bleak and hopeless to me.	Feel desperate.
I rarely feel lonely or blue. [R]	Have frequent mood swings.
Sometimes I feel completely worthless.	Am often down in the dumps.
I have sometimes experienced a deep sense of guilt or sinfulness.	Feel comfortable with myself. [R]
I tend to blame myself when anything goes wrong.	Dislike myself.
Too often, when things go wrong, I get discouraged and feel like giving up.	Feel that my life lacks direction.
	Often feel blue.
	Am very pleased with myself. [R]

Self-consciousness (N4) **(alpha = 0.76, omega = 0.77)**	**Self-consciousness (N4)** **(alpha = 0.81, omega = 0.82)**
It doesn't embarrass me too much if people ridicule and tease me. [R]	Am not embarrassed easily. [R]
I often feel inferior to others.	Am easily intimidated.
In dealing with other people, I always dread making a social blunder.	Am afraid that I will do the wrong thing.
I seldom feel self-conscious when I'm around people. [R]	Am able to stand up for myself. [R]
At times I have been so ashamed I just wanted to hide. [R]	Stumble over my words.
I feel comfortable in the presence of my bosses or other authorities. [R]	Find it difficult to approach others.
If I have said or done the wrong thing to someone, I can hardly bear to face them again.	Am afraid to draw attention to myself.
When people I know do foolish things, I get embarrassed for them.	Only feel comfortable with friends.
	Am comfortable in unfamiliar situations. [R]
	Am not bothered by difficult social situations. [R]

Impulsiveness (N5) **(alpha = 0.74, omega = 0.74)**	**Immoderation (N5)** **(alpha = 0.79, omega = 0.79)**
I have trouble resisting my cravings.	Am able to control my cravings. [R]
I have little difficulty resisting temptation. [R]	Easily resist temptations. [R]
I rarely overindulge in anything. [R]	Rarely overindulge. [R]
When I am having my favorite foods, I tend to eat too much.	Often eat too much.
Sometimes I do things on impulse that I later regret.	Do things that I later regret.
I sometimes eat myself sick.	Never splurge. [R]
I am always able to keep my feelings under control. [R]	Don't know why I do some of the things I do.
I seldom give in to my impulses. [R]	Love to eat.
	Go on binges.
	Never spend more than I can afford. [R]

Vulnerability (N6) **(alpha = 0.78, omega = 0.78)**	**Vulnerability (N6)** **(alpha = 0.83, omega = 0.83)**
I'm pretty stable emotionally. [R]	Get overwhelmed by emotions.
I feel I am capable of coping with most of my problems. [R]	Know how to cope. [R]
I keep a cool head in emergencies. [R]	Am calm even in tense situations. [R]
I often feel helpless and want someone else to solve problems.	Can handle complex problems. [R]
It's often hard for me to make up my mind.	Can't make up my mind.
I can handle myself pretty well in a crisis. [R]	Panic easily.
When everything seems to be going wrong, I can still make good decisions. [R]	Readily overcome setbacks. [R]
When I'm under a great deal of stress, sometimes I feel like I'm going to pieces	Feel that I'm unable to deal with things.
	Become overwhelmed by events.
	Remain calm under pressure. [R]

Note. Reverse-items are denoted by [R].

Table 1

Whole-Sample Networks: Descriptive Overview of Network Characteristics

		NEO	IPIP
Network characteristic		8-items	10-items
Global strength		2.41	2.40
Average edge strength: M (SD)		0.24 (0.08)	0.30 (0.08)
Number of non-zero edges		10	8
Number of zero edges		5	7
Network density		0.67	0.53
Largest edge weight		DEP–SELF (0.35)	ANX–ANG (0.44)
Strength of most different edge (ANX–ANG)		0.16	0.44
CS-coefficient for Expected Influence [a]		0.75	0.75
Centrality ranks (Expected Influence raw score)			
	1st	DEP (1.17)	ANX (1.15)
	2nd	VUL (0.91)	DEP (1.12)
	3rd	ANX (0.89)	VUL (0.85)
	4th	ANG (0.70)	ANG (0.65)
	5th	SELF (0.68)	SELF (0.58)
	6th	IMP (0.48)	IMP (0.45)

Note. All values were computed on the full sample of $N = 424$. ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability.
a. *CS*-coefficients are an index of centrality stability, and represent the maximum proportion of sampled cases that can be dropped in order to retain a correlation of 0.7 (with 95% probability) between original sample and subsample centrality estimates (Epskamp et al., 2018).

Table 2

Scale Variability Condition: Descriptive Overview of Network Characteristics

Network characteristic Mean (SD)	1-item		3-items		8-items	
	NEO	IPIP	NEO	IPIP	NEO	IPIP
Average global strength	1.50 (0.24)	1.59 (0.20)	2.10 (0.13)	2.12 (0.12)	2.39 (0.05)	2.39 (0.10)
Average edge strength [a]	0.24 (0.03)	0.24 (0.02)	0.26 (0.02)	0.27 (0.03)	0.27 (0.02)	0.28 (0.03)
Mean no. of non-zero edges (out of a possible 15)	6.38 (1.31)	6.80 (1.09)	8.02 (1.02)	8.04 (0.97)	8.90 (0.68)	8.62 (0.92)
Mean no. of zero edges (out of a possible 15)	8.62 (1.31)	8.20 (1.09)	6.98 (1.02)	6.96 (0.97)	6.10 (0.68)	6.38 (0.92)
Avg. network density	0.43	0.45	0.53	0.54	0.59	0.57
Avg. correlation between edge weights	r_p = .25 (.23)		r_p = .47 (.17)		r_p = .66 (.09)	
Avg. correlation between centrality scores	r_s = .47 (.36)		r_s = .70 (.24)		r_s = .89 (.10)	
Centrality ranks: Expected Influence (frequencies, out of 50, listed in parentheses)						
Most frequent central node (i.e., ranked 1st most often)	DEP (29)	DEP (23)	DEP (36)	DEP (32)	DEP (49)	DEP (31)
Second most frequent central node (i.e., ranked 1st second most often)	ANX (10)	ANX (13)	VUL (8)	ANX (13)	VUL (1)	ANX (17)
Third most frequent central node (i.e., ranked 1st third most often)	SELF (6)	VUL (9)	ANX (5)	VUL (5)	n/a	VUL (2)
Largest edge weights (frequencies, out of 50, listed in parentheses)						
Most frequent largest edge	ANX–DEP (15)	ANG–DEP DEP–VUL (9)	DEP–VUL (19)	ANX–ANG (12)	DEP–VUL (26)	ANX–ANG (35)

Second most frequent largest edge	DEP–SELF (9)	ANX–ANG (8)	ANX–DEP (13)	DEP–VUL (10)	DEP–SELF (22)	ANX–DEP ANX–VUL (6)
Third most frequent largest edge	ANG–DEP (6)	ANX–VUL (7)	DEP–SELF (12)	ANX–DEP (8)	ANX–DEP (2)	DEP–SELF (2)
Maximum edge differences (frequencies, out of 50, listed in parentheses)						
Most frequent max edge discrepancy	ANX–ANG (7)		ANX–ANG (16)		ANX–ANG (15)	
Second most frequent max edge discrepancy	ANG–DEP DEP–SELF ANX–VUL DEP–VUL (6)		ANG–IMP SELF–VUL (6)		ANX–SELF (10)	
Third most frequent max edge discrepancy	ANX–DEP (4)		DEP–SELF (5)		DEP–IMP (6)	

Note. All values were computed on a sample of n = 212. ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability. r_p and r_s denote Pearson and Spearman-rank correlation coefficients, respectively.

a. Average edge strength was computed as the mean of absolute edge weights across all non-zero edges in the network.

Table 3

Whole-Sample Networks: Edge Weights

	NEO	IPIP
ANX–ANG	0.16	0.44
ANX–DEP	0.26	0.35
ANX–SELF	0.20	-
ANX–IMP	-	-
ANX–VUL	0.27	0.36
ANG–DEP	0.23	-
ANG–SELF	-	-
ANG–IMP	0.31	0.21
ANG–VUL	-	-
DEP–SELF	0.35	0.31
DEP–IMP	-	0.24
DEP–VUL	0.34	0.22
SELF–IMP	-	-
SELF–VUL	0.13	0.28
IMP–VUL	0.17	-

Note. All values were computed on the full sample of N = 424. ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability. Only edge weights estimated as non-zero are provided above.

Figure 1

Differences in Adjacency Matrices: Edges Estimated as Present versus Absent

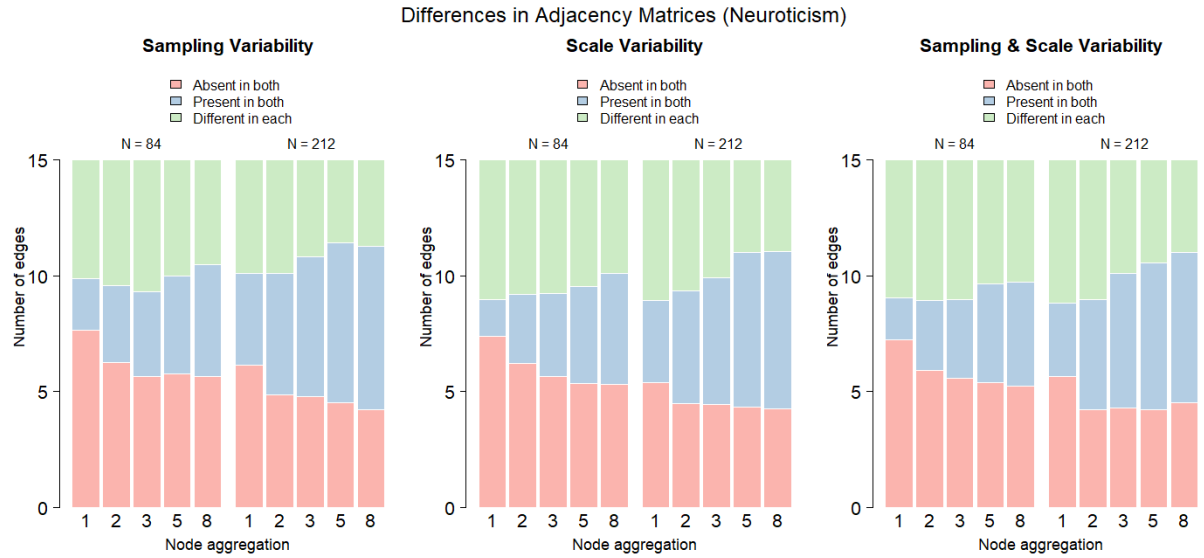


Figure 2

Average Global Strength

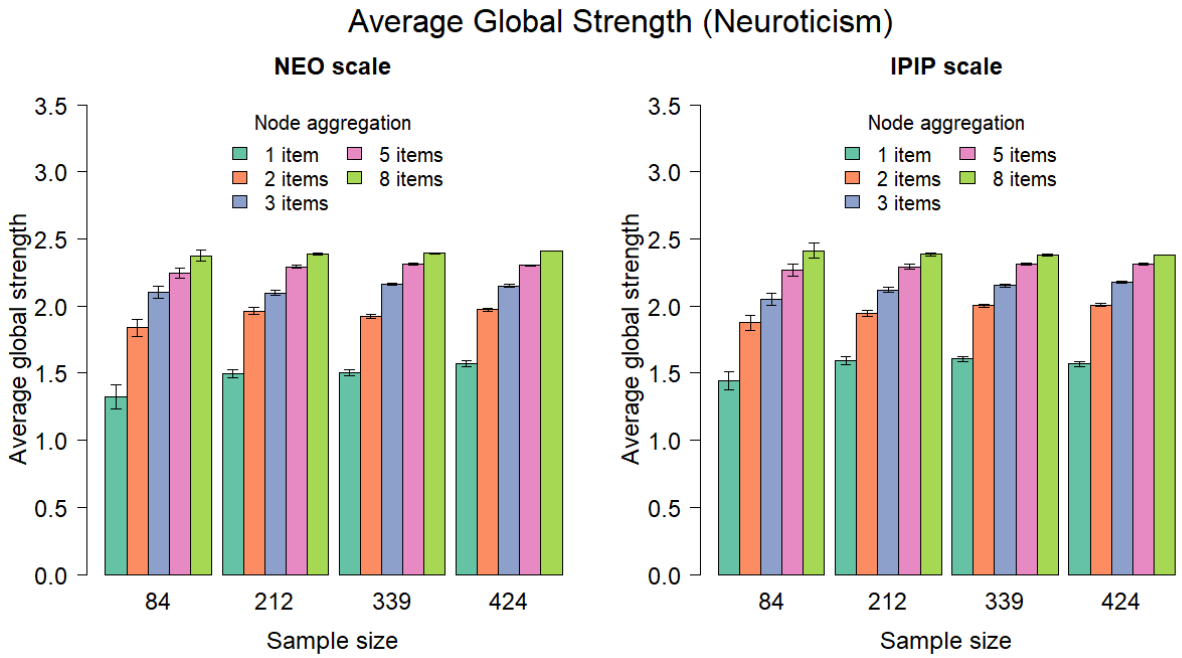


Figure 3

Differences in Average Global Strength

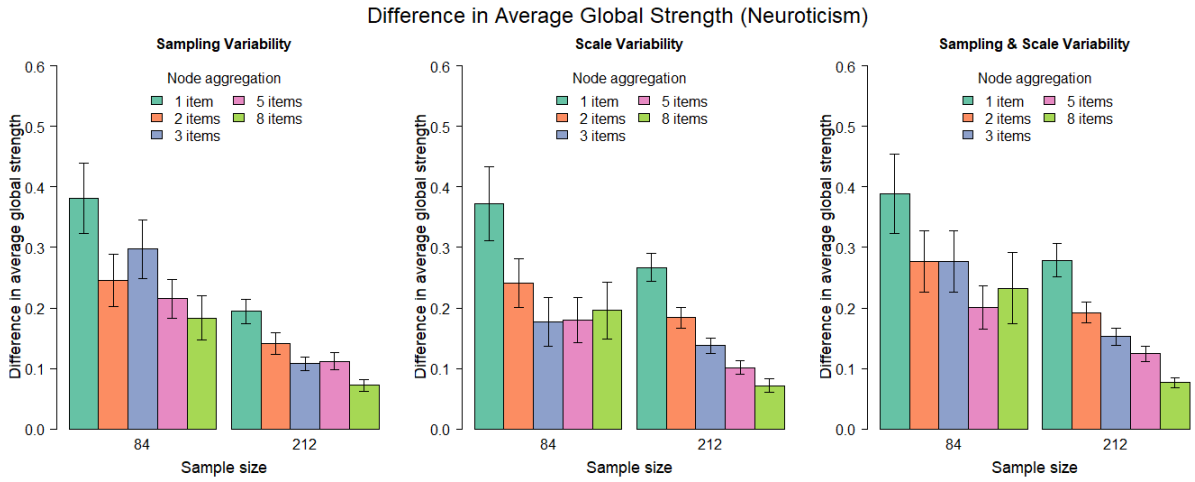


Figure 4

Correlations Between Edge Weights

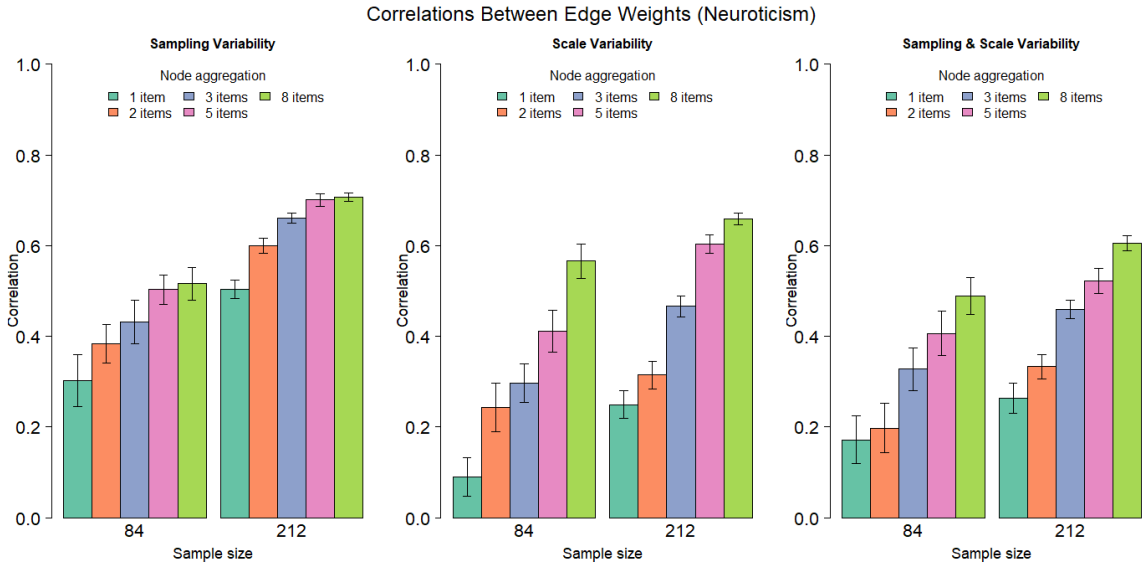


Figure 5

Correlations Between Centrality Scores (Expected Influence)

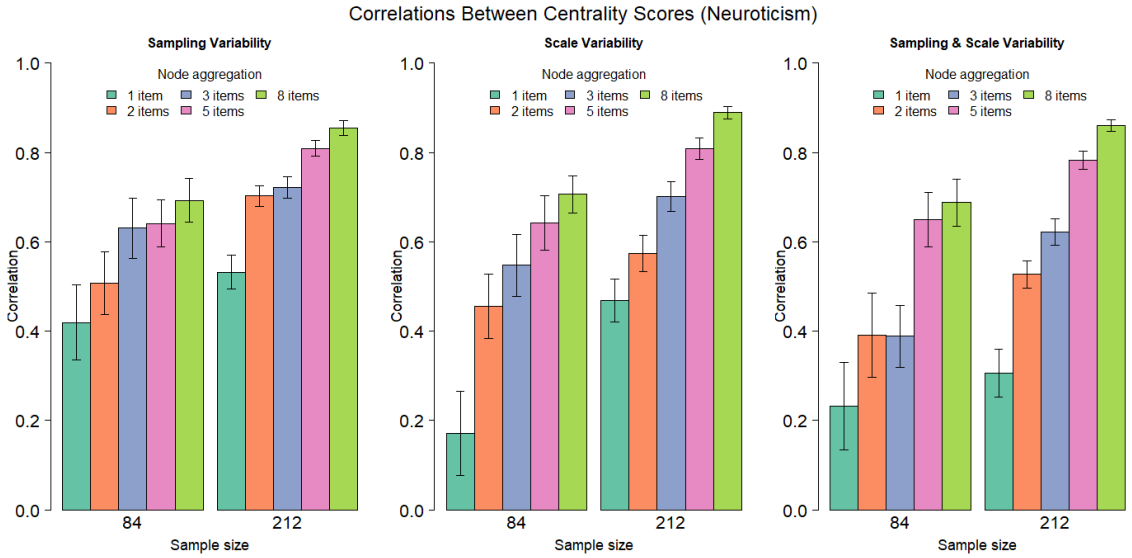


Figure 6

Whole-Sample Networks

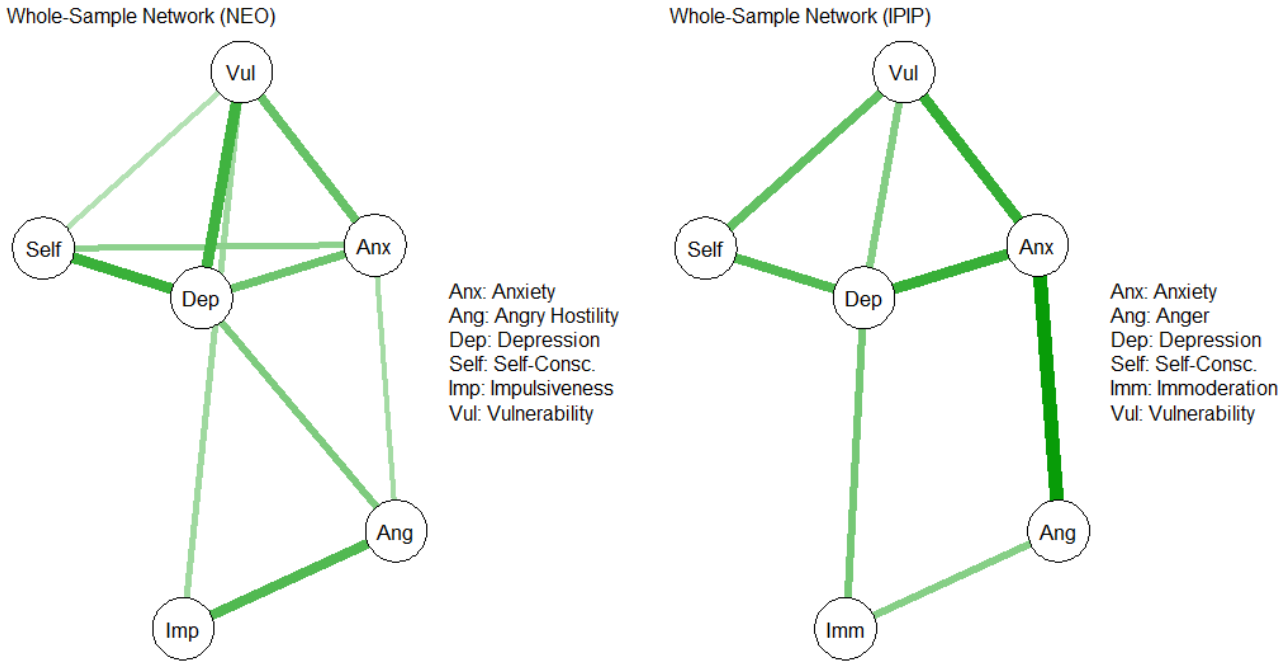
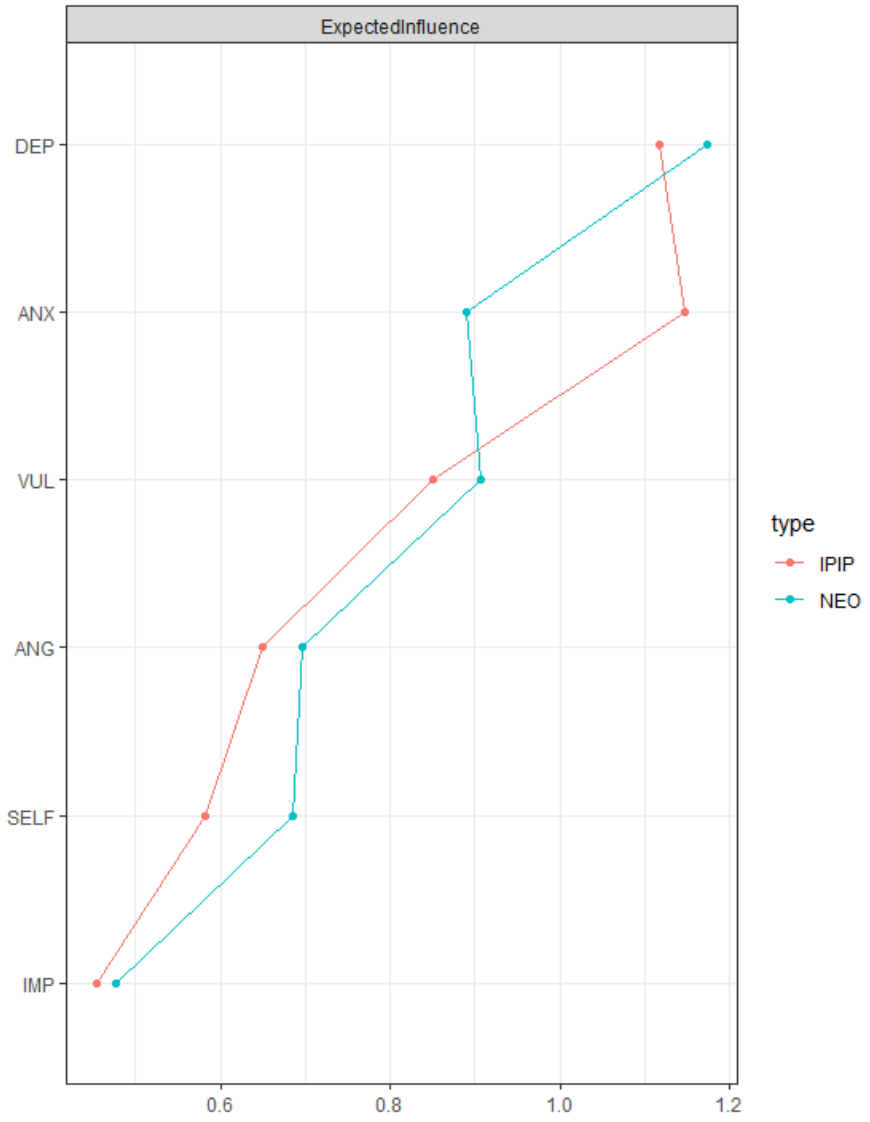


Figure 7

Whole-Sample Networks: Correspondence in Expected Influence Scores



List of Figure Captions

Figure 1	*Note.* Effects of sample size and node aggregation on differences in adjacency matrices, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Results are displayed in the form of stacked bar charts, such that each bar height represents the total number of possible network edges (i.e., 15), and coloured bar segments break down the relative proportions of total edges that were consistently estimated as absent (red bars) or present (blue bars), or different (green bars).
Figure 2	*Note.* Effects of sample size and node aggregation on average global strength of networks. Neuroticism networks were estimated via use of the NEO scale (left) and the IPIP scale (right). Error bars represent 95% confidence intervals around the mean.
Figure 3	*Note.* Effects of sample size and node aggregation on differences in average global strength, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.
Figure 4	*Note.* Effects of sample size and node aggregation on correlations between edge weights, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.
Figure 5	*Note.* Effects of sample size and node aggregation on rank correlations between centrality (i.e., Expected Influence) scores, across resampling conditions: sampling variability (left), scale variability (middle), sampling and scale variability (right). Error bars represent 95% confidence intervals around the mean.
Figure 6	*Note.* Non-regularized GGM networks for the Neuroticism network, estimated on the full sample (N = 424), using the NEO scale (left) vs. the IPIP scale (right). Solid-green lines represent positive edge weights; dashed-red lines represent negative edge weights. Networks were plotted using the average of the two Fruchterman-Reingold (or "spring") layouts in *qgraph* (max edge weight set to 0.45; Epskamp et

	al., 2012). Largest edge weights for the NEO and IPIP networks are, resp., between DEP and SELF (edge weight = 0.35), and ANX and ANG (edge weight = 0.44).
Figure 7	*Note.* Expected Influence scores for whole-sample networks are plotted in terms of raw values along the x-axis, with nodes ordered by (descending) rank along the y-axis. Blue line represents estimates derived from the NEO scale, and red line represents estimates derived from the IPIP scale. All values were computed on the same sample of subjects (i.e. full sample of *N* = 424). ANX = anxiety, ANG = anger, DEP = depression, SELF = self-consciousness, IMP = impulsiveness, and VUL = vulnerability. Most central node, as estimated by the NEO and IPIP scales, respectively, are Depression and Anxiety.